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Prescription Model for Variable Rate Adaptive Sprayer System

BE ACCEPTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE
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**MASTER OF SCIENCE IN
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Prescription Model for Variable Rate Adaptive Sprayer System

By

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Abstract

The growing demand for sustainable farming practices has accelerated the development of precision agriculture technologies that reduce waste while improving crop production. This capstone extends prior research on a modular precision sprayer system for the Agbot rover by advancing nozzle-level control and prescription-based operation. A 1-to-1 architecture, in which each nozzle is supplied by a dedicated pump, is implemented and validated. Laboratory tests achieved steady-state errors of 5.3 - 6.1%, while field tests resulted in errors ranging from 9.1 - 28.9%.

Furthermore, a prescription model was developed to integrate rover position, speed, and row width into nozzle flow commands at varying degrees of complexity. Field validation using volume-distance plots confirmed that prescribed application rates were met across multiple zones, with average root mean square (RMS) errors ranging from 0.9 - 32.2%. These results represent a step toward scalable, prescription-driven spraying for autonomous agricultural robotics.

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Contents

Abstract	iii
Acknowledgments	iv
Contents	v
List of Figures	vi
List of Tables	vii
1 Introduction	1
1.1 Background and Motivation	1
1.2 Ethical Implications and Considerations	2
1.3 Contributions	2
2 System Overview	4
2.1 Mechanical Subsystem	5
2.1.1 Fluid Architecture	5
2.2 Electrical Subsystem	6
2.3 Software Subsystem	6
2.4 Sprayer and Agbot Integration	7
3 Control and Validation	9
3.1 Banked Architecture	9
3.1.1 Banked Mapping Function	12
3.1.2 Testing and Results	12
3.2 1-to-1 Architecture	14
3.2.1 1-to-1 Mapping Function	15
3.2.2 Testing and Results	16
4 Prescription Model	21
4.1 Prescription Model Validation	23
5 Conclusion and Future Work	25
5.1 Conclusion	25
5.2 Future Work	25

List of Figures

2.1	Agbot Rover and Sprayer System in the field	4
2.2	Sprayer System Overview	5
2.3	Sprayer System Fluid Subsystem	6
2.4	Sprayer System Wiring Diagram	7
2.5	Sprayer System Component Block Diagram	8
3.1	Control Block Diagram Representation of Sprayer Banked Control System	10
3.2	Left-bank open loop response at different pump powers and valve positions.	13
3.3	Right-bank open loop response at different pump powers and valve positions.	13
3.4	In-Lab Multi-Zone Time Response for banked architecture. The top plot shows the flow rate response to desired step flow rate inputs. The middle plot shows the pump powers computed by the closed-loop pump controller. The bottom plot shows the valve positions computed by the closed-loop valve controller. Note: Pumps 1 and 3 are inactive in the banked system.	14
3.5	Control Block Diagram Representation of Sprayer 1-to-1 Control System	16
3.6	Sprayer system open loop response for each nozzle unit in 1-to-1 architecture. Each valve is set to 80% open.	17
3.7	In-Lab Multi-Zone Time Response for 1-to-1 architecture. The top plot shows the flow rate response to desired step flow rate inputs. The bottom plot shows the pump powers computed by the closed-loop controller.	18
3.8	Multi-Zone Time Response in the field. The top plot shows the flow rate response to desired step flow rate inputs. The bottom plot shows the pump powers computed by the closed-loop controller.	19
4.1	Prescription Model Levels of Complexity. (a) Basic prescription with constant volume per area. (b) Simple prescription with different volumes per area based on longitudinal position. (c) Complex prescription with different volumes per area longitudinally and laterally.	22
4.2	Instantaneous volume dispensed as the Agbot traverses 80 meters. Top plot shows volume dispensed in the field. Bottom plot shows desired volume dispensed.	24
4.3	Cumulative volume dispensed by each nozzle as Agbot travels 80 meters.	24

List of Tables

3.1	Banked Response Characterization	15
3.2	Nozzle Sensor Accuracy and Validation	17
3.3	Desired Flow Rates for Multi-Zone Test	18
3.4	In Lab Multi Zone Response	19
3.5	Field Multi Zone Response	20
4.1	Prescription Model Response: Average Root Mean Square (RMS) Error (%)	23

Chapter 1

Introduction

1.1 Background and Motivation

The global demand for higher agricultural yields and more efficient resource use has led to an increase in demand for precision agriculture technologies. Conventional sprayer systems uniformly apply volumes of pesticides, fertilizers, or irrigation across entire fields regardless of crop density, pest distribution, or soil condition [1]. This blanket application strategy often results in wasted resources and environmental harm.

Conventional spraying systems typically include tractor-mounted boom sprayers and orchard air-blast sprayers, which deliver chemicals at a fixed rate across entire fields. While simple and robust, these systems often result in chemical drift, non-uniform coverage, and high energy consumption, particularly when crop canopy density varies [1] [2]. Modern sprayer designs have introduced partial solutions. Additionally, conventional irrigation methods such as flood irrigation, furrow irrigation, and fixed sprinkler systems distribute large volumes of water evenly, without accounting for spatial variability [3] [4].

Precision agriculture allows for the optimization of land, water, fuel, fertilizer, and pesticide use. Thus, enabling farmers to be more sustainable and efficient with their resource use, which reduces both cost and environmental impact. The primary areas of precision agriculture being explored are:

1. Auto guidance of machinery to aid in tilling, planting, spraying, and harvesting
2. Machine section control to optimize material application and planting
3. Variable rate technology (VRT) to determine the application rates of fertilizer, seed, and crop protection products
4. Machine and fleet analytics to increase asset utilization and decrease the amount of fuel used
5. Precision irrigation to reduce water consumption without negatively impacting crop performance [5].

Through VRT, farmers are able to vary the application rate of fertilizers, pesticides, and water to maximize crop output and minimize resource waste. Variable rate technology can utilize either GPS-guided prescription maps or real-time sensors to identify zones

that require more or less resources. Tailoring the distribution of resources to specific field zones boosts crop yields and preserves soil health. Variable rate technology can be used in conjunction with precision irrigation to provide crops with the necessary moisture without risking fungal diseases or root rot that can be caused by over-watering [6]. Central to the precision agriculture approach are prescription models, or mathematical frameworks that determine how much input should be applied in each location and when.

Furthermore, an emerging trend in agriculture is the desire to plant multiple crops within the same row, a strategy that increases biodiversity, improves resilience, and reduces the risk of total yield loss if one crop is damaged by pests or disease. However, traditional irrigation systems limit this practice because all plants in a row typically receive the same amount of water regardless of species-specific needs. Prescription irrigation technologies can overcome this challenge by delivering precise volumes of water to different zones or even to specific plant rows, thereby enabling multi-crop planting strategies that enhance both sustainability and farmer profitability [7].

1.2 Ethical Implications and Considerations

Chemical exposure is a well-documented occupational hazard for farm workers and the nearby communities. Variable-rate and precision application systems can help lower exposure risks by allowing a robotic system to supplant a farmer in this task. The automated application of chemicals and pesticides can also mitigate chemical overuse and protect biodiversity in the surrounding environment [5]. However, the dependence on automation must also be considered. If the automated systems fail or are misused, they may inadvertently increase exposure risk through over-spraying [8]. Additionally, the socio-technical impacts of automation, specifically the future of farm labor, must be considered when designing and developing agricultural technologies. While automated sprayers reduce the need for manual spraying tasks and improve safety, they also contribute to labor displacement. And thus, the efficiency gained through technology must be balanced with social responsibility by supporting workforce transitions through worker retraining or complementary roles, allowing humans and robots to co-exist in the field [9].

1.3 Contributions

The scope of the presented work is to build on the modular precision sprayer system developed by Santa Clara University's Robotics Systems Laboratory. This sprayer system has been previously integrated into an existing agricultural rover known as the Agbot [10]. The system created by Zhuta et al. lacked nozzle-level precision and a fully integrated prescription model.

This research investigates

1. independent nozzle control for a four-nozzle sprayer system,
2. banked and 1-to-1 control architectures, and

3. the development of a prescription model that determines nozzle flow rates based on rover speed, position, and row width.

This integration of hardware and control algorithms represents a step toward fully automated, data-driven precision spraying systems, to support the goal of sustainable and efficient agriculture.

Chapter 2

System Overview

The modular sprayer system (Figure 2.1 and 2.2) is designed to be a tool add-on to the AgBot rover. The initial sprayer system used in this research was previously developed by L. Zhuta, et al. [10]. The following sections describe the hardware and software architecture that compose the sprayer system and highlights the changes made from the previous version.

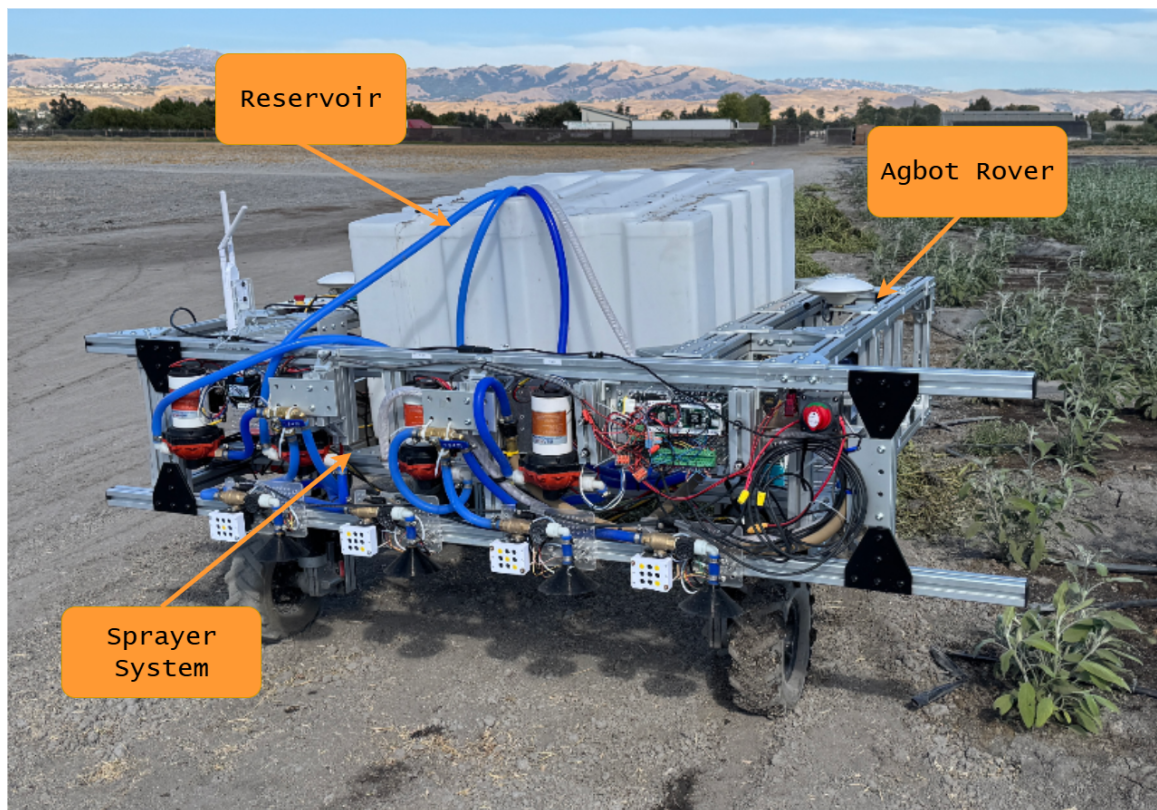


Figure 2.1 – Agbot Rover and Sprayer System in the field

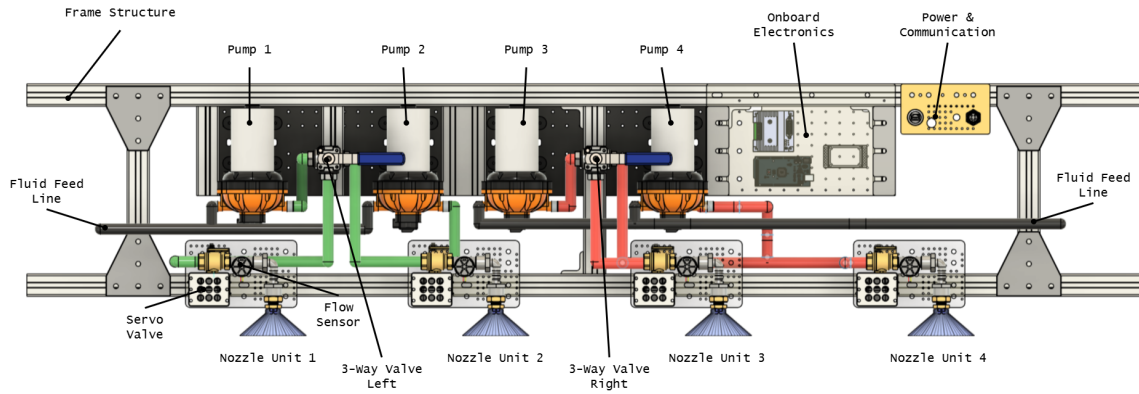


Figure 2.2 – Sprayer System Overview

2.1 Mechanical Subsystem

The frame of the sprayer is constructed out of aluminum T-slot rails, allowing it to be easily mounted onto the AgBot. The aluminum T-slot rails also allows for the sprayer system to be easily rearranged or modified. Potential modifications include changing the number of pumps or nozzle units and adjusting the distance between nozzle units.

2.1.1 Fluid Architecture

Attached to the sprayer's frame is the fluid subsystem. This subsystem consists of a fluid reservoir, four positive-displacement pumps, four modular nozzle units (NUs), and silicone tubing. Each NU contains a servo valve, a flow sensor, and a nozzle.

Two different flow configurations are explored: a banked system and a 1-to-1 system. The banked system utilizes two pumps to power all four nozzles, creating a coupled system in which the output flows of the NUs on one bank are related. The 1-to-1 system has four pumps, allowing each NU to be independently supplied with fluid, thus allowing for simpler independent control of all four NUs. In order to easily switch between configurations, manual 3-way valves were added to switch which pump a nozzle was fed fluid from. Figure 2.3 shows the flow of fluid through the system, from the reservoir to the outlet of each nozzle.

The banked approach offers advantages in terms of hardware simplicity, cost, and robustness. With fewer pumps and control channels, the overall system is lighter and easier to maintain. However, with this approach, the nozzle outputs within the same bank are inherently coupled, and achieving truly independent control becomes challenging. On the other hand, the 1-to-1 approach enables each nozzle to be controlled independently with finer spatial resolution. The tradeoff is that this design requires more pumps, wiring, and power, which ultimately increases system complexity and cost, and may introduce additional points of failure in field conditions. Balancing these competing considerations of system complexity, control precision, cost, and adaptability motivated the evaluation of both architectures within the modular sprayer platform.

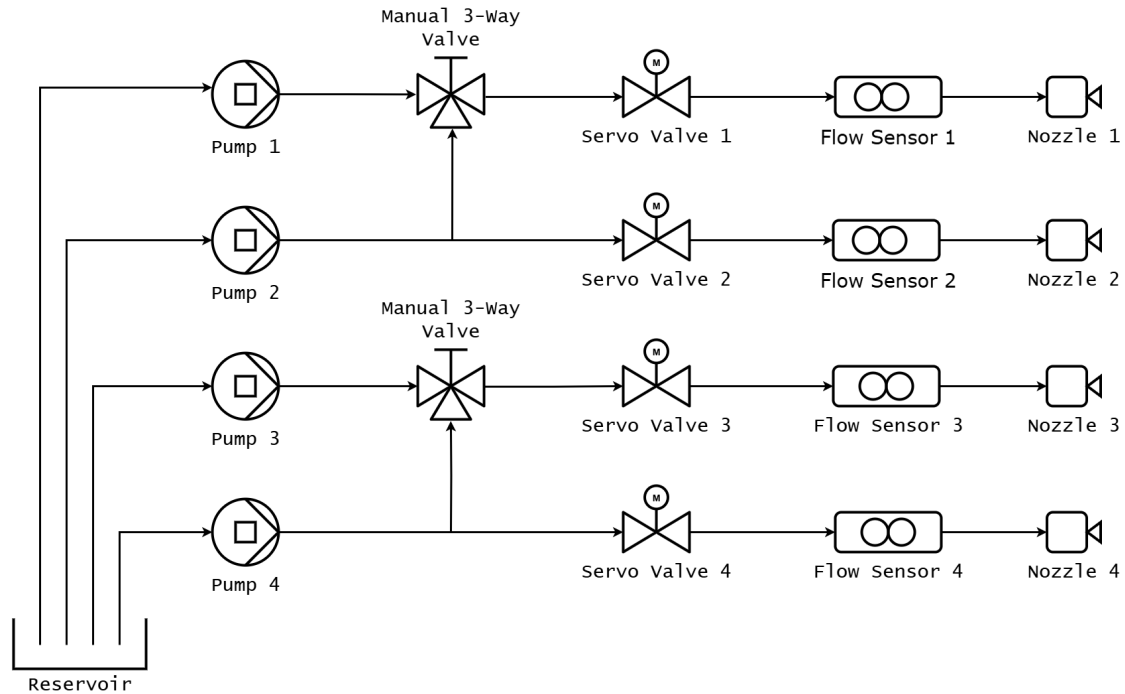


Figure 2.3 – Sprayer System Fluid Subsystem

2.2 Electrical Subsystem

The sprayer utilizes an Arduino Mega 2560 micro-controller to control the pump power and servo-valve position based on instantaneous flow rate information. The flow rate information is computed by an Arduino Uno that reads pulse counts from the flow sensors in each nozzle unit. Based on these pulse counts, the Arduino Uno calculates the flow rate and applies a moving-average filter before sending this information to the main Arduino Mega controller. The pumps are controlled using RoboteQ motor controllers.

On the main AgBot, a Jetson Nano is utilized to exchange information between the sprayer system and the rover. The sprayer is designed to operate at 12 VDC and is powered by a 24 V battery separate from the AgBot's power supply. Figure 2.4 depicts the electrical system of the sprayer system.

2.3 Software Subsystem

An error-based proportional-integral (PI) controller with a feed-forward term is used, and is based on the desired and measured flow rates of the sprayer system. Real-time flow rate and pump power data is recorded for each nozzle unit. In the lab, this data can be displayed in the serial monitor within the Arduino IDE or collected in CSV files using MATLAB. In the field, this data is stored in ROS 2 bags.

The sprayer system communicates using Serial. The sensing Arduino Uno sends measured flow rate information over Serial to the controller Arduino Mega 2560. Additionally, the Arduino Mega 2560 communicates with the RoboteQ and the Jetson Nano using Serial.

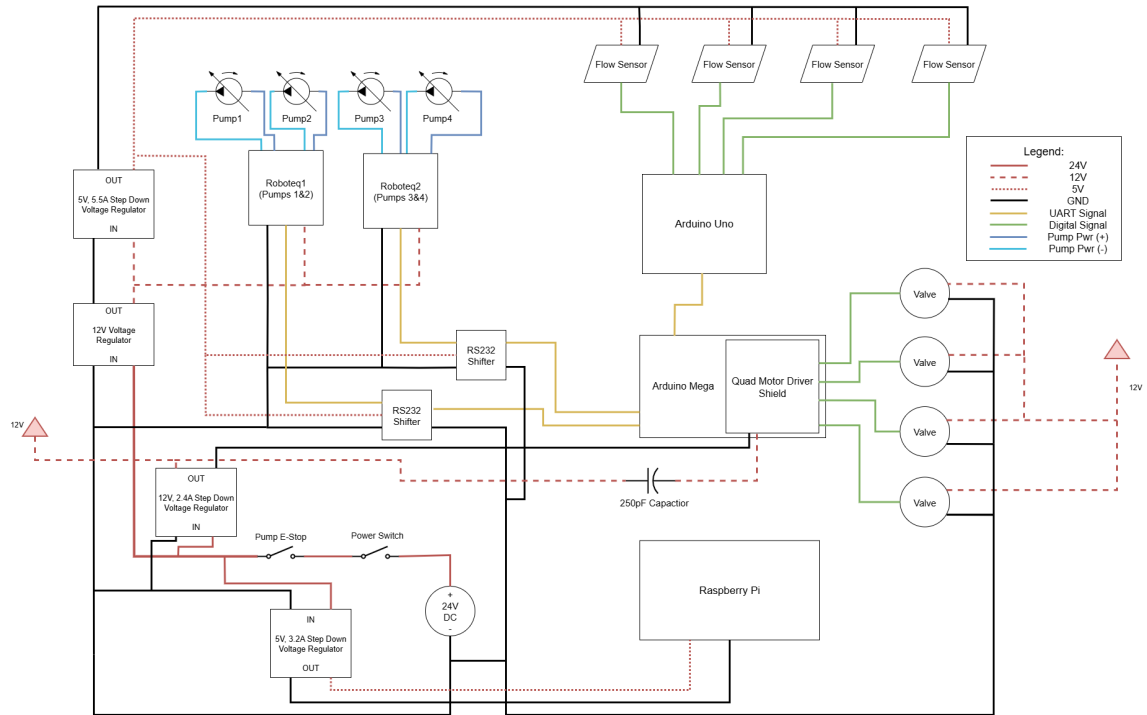


Figure 2.4 – Sprayer System Wiring Diagram

2.4 Sprayer and Agbot Integration

The navigation system, a key component for the integration of the sprayer system, is responsible for maintaining the rover on a predefined path and maintaining a desired velocity along said path. The system uses real time RTK-GPS position and velocity, allowing the rover to maintain heading, position, and velocity at centimeter-level accuracy. Paths are defined as a composition of line, arc, and Catmull-Rom spline segments. Each segment is associated with a segment ID, allowing us to identify which segment the rover is currently on.

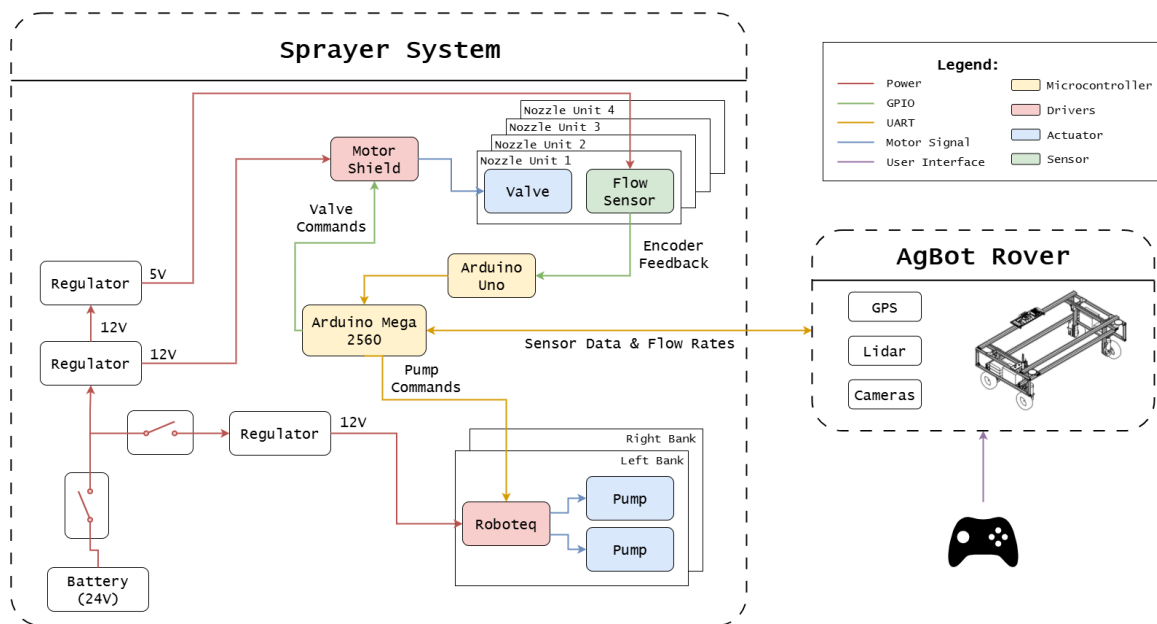


Figure 2.5 – Sprayer System Component Block Diagram

Chapter 3

Control and Validation

The control algorithm of the sprayer system is designed to achieve accurate and stable independent nozzle control. Each nozzle must be able to track a command flow rate within a $\pm 10\%$ error margin, even when operating alongside other nozzles or under varying environmental conditions.

In this research, a proportional-integral (PI) controller with a look-up table mapping function is utilized. While Proportional-Integral-Derivative (PID) controllers are commonly used robust control methods, a PI controller with a feed-forward term was chosen as the control scheme because of its ability to preemptively determine the control effort needed based on the open-loop behavior of the system.

Since there are two different flow architectures, two different control schemes were implemented. Both architectures have the same structure of a PI controller with a feed-forward mapping function. The mapping function takes the form of a look-up table.

The inputs for each controller are the desired flow rates of each nozzle unit, $\bar{q}_1$, $\bar{q}_2$, $\bar{q}_3$, and $\bar{q}_4$. The mapping function takes these desired flow rates as inputs and converts them to open-loop pump powers. Then, based on the resultant error, the PI controller adjusts the pump power value to minimize steady-state error. The flow rate of each nozzle unit is measured by a flow sensor that sits between the valve and nozzle on each NU.

3.1 Banked Architecture

The banked architecture supplies pairs of nozzles using a shared pump, creating a coupled system in which the output of one nozzle is influenced by the demand of its partner. This design reduces hardware complexity and cost, but introduces challenges for achieving independent nozzle control.

The banked system utilizes pumps 2 and 4 to power the left and right banks, respectively. The mapping function determines the desired flow rates for each bank ($\bar{q}_1$ and $\bar{q}_2$ or $\bar{q}_3$ and $\bar{q}_4$) based on the current valve positions (V_1 , V_2 , V_3 , and V_4) and pump power (u_l and u_r).

$$M_l(V_1, V_2, \hat{u}_l) \rightarrow q_1, q_2$$

$$M_r(V_3, V_4, \hat{u}_r) \rightarrow q_3, q_4$$

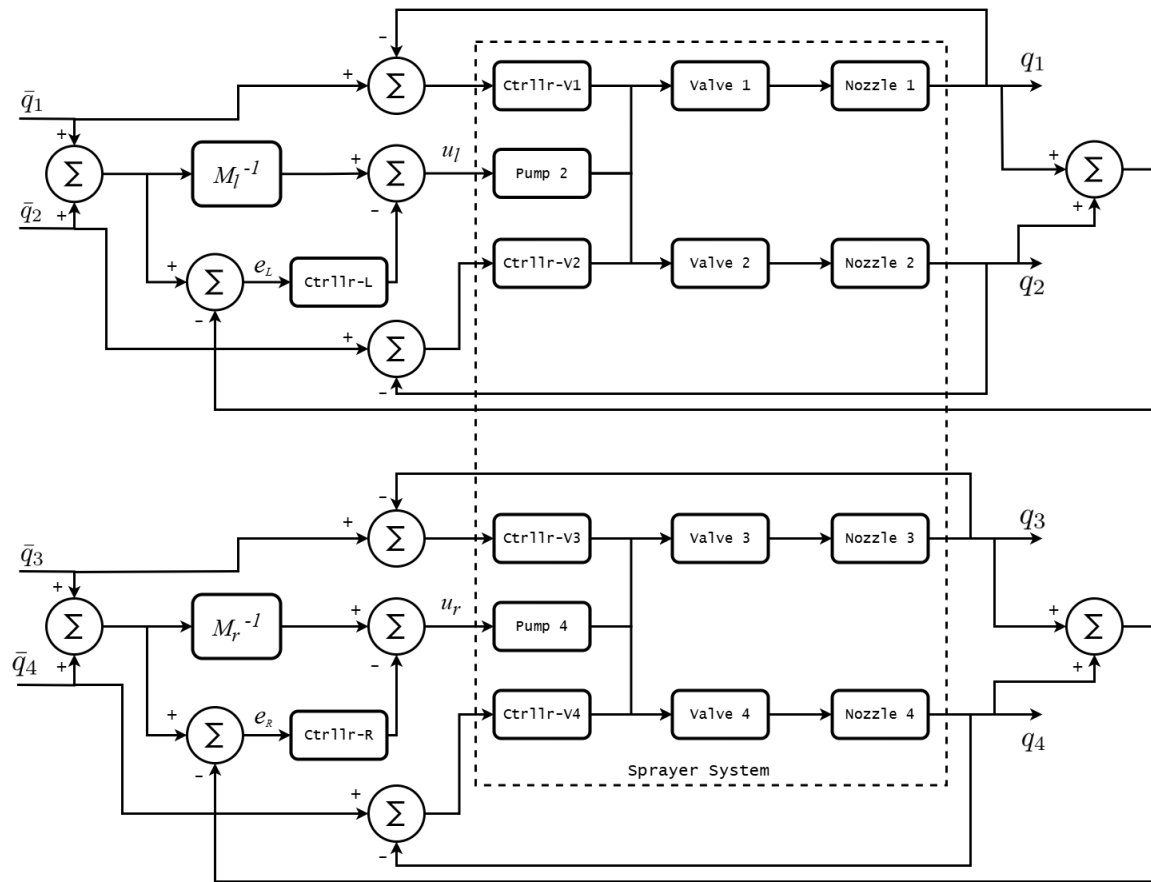


Figure 3.1 – Control Block Diagram Representation of Sprayer Banked Control System

The inverse mapping function is utilized to calculate the open-loop pump power for the given desired flow rates. Based on these desired flow rates, the respective errors for each bank are

$$e_l = (\bar{q}_1 + \bar{q}_2) - (q_1 + q_2)$$

$$e_r = (\bar{q}_3 + \bar{q}_4) - (q_1 + q_2)$$

Using the calculated errors, the pump powers for the left and right banks are

$$u_l = \hat{u}_l + (K_{p_l} e_l + K_{i_l} \int e_l)$$

$$u_r = \hat{u}_r + (K_{p_r} e_r + K_{i_r} \int e_r)$$

where K_{p_l} and K_{p_r} are the proportional terms of the banked controller and K_{i_l} and K_{i_r} are the integral terms.

In order to get independent nozzle control of the banked system, an additional PI controller is implemented to control valve position. The valve PI controllers utilize the error of each individual nozzle

$$e_j = \bar{q}_j - q_j$$

$$j = 1 \dots 4$$

The valve in the nozzle unit with the highest desired flow rate will be set to all the way open, and the PI gains set to zero, effectively turning off the PI controller for that valve. The valve position of the other nozzle unit is set based on the ratio between the desired flow rates.

$$R_l = \frac{\bar{q}_1}{\bar{q}_2}$$

$$R_r = \frac{\bar{q}_3}{\bar{q}_4}$$

If $R_l > 1.0$, then

$$V_1 = 100$$

$$\hat{V}_2 = \frac{V_1}{R_l}$$

$$V_2 = \hat{V}_2 + (K_{p_2} e_2 + K_{i_2} \int e_2)$$

and $K_{p_1} = 0$ and $K_{i_1} = 0$.

Otherwise,

$$V_2 = 100$$

$$\hat{V}_1 = V_2 * R_l$$

$$V_1 = \hat{V}_1 + (K_{p_1} e_1 + K_{i_1} \int e_1)$$

and $K_{p_2} = 0$ and $K_{i_2} = 0$.

If $R_r > 1.0$, then

$$V_3 = 100$$

$$\hat{V}_4 = \frac{V_3}{R_r}$$

$$V_4 = \hat{V}_4 + (K_{p4}e_4 + K_{i4} \int e_4)$$

and $K_{p3} = 0$ and $K_{i3} = 0$.

Otherwise,

$$V_4 = 100$$

$$\hat{V}_3 = V_4 * R_r$$

$$V_3 = \hat{V}_3 + (K_{p3}e_3 + K_{i3} \int e_3)$$

and $K_{p4} = 0$ and $K_{i4} = 0$.

3.1.1 Banked Mapping Function

In order to fully understand the coupling in each bank, open-loop system tests were run for each bank. Figures 3.2 and 3.3 show the open-loop response for each nozzle unit at varying pump powers and valve positions. They display the effect of opening or closing one valve on the other nozzle unit in the same bank. Based on this data, 3D look-up tables were created to act as the mapping functions. Each look-up table takes three inputs: a pump power and two valve positions. Different combinations of pump power and valve positions correspond to a flow rate for the nozzle.

3.1.2 Testing and Results

The in-lab testing of the banked architecture revealed fundamental challenges with achieving independent nozzle control due to the coupled nature of the system. The multi-zone time response (Figure 3.4) highlights the limitations of this configuration. While the pump PI controllers adjusted effort in response to bank-level error signals, the valve controllers were unable to fully compensate for nozzle-to-nozzle discrepancies. As a result, the nozzle with the higher flow demand dominated the bank's performance, and the paired nozzle consistently exhibited steady-state overshoot. This imbalance persisted even when PI gains were tuned aggressively, suggesting that the underlying fluid dynamics of the banked architecture fundamentally constrain independent nozzle control.

In addition, Table 3.1 illustrates a relationship between the difference in desired flow rates ($\Delta\bar{q}$) within a bank and the resulting steady-state error. At low $\Delta\bar{q}$ values (0-10L/h), the system maintains moderate errors of roughly 8-15%. However, as $\Delta\bar{q}$ increase to 30-50L/h, the steady-state error increases drastically. This behavior confirms that the banked architecture cannot sustain accurate independent control when nozzle setpoints diverge significantly.

Overall, while the banked configuration reduces hardware costs, it cannot reliably achieve independent nozzle control within a $\pm 10\%$ error margin. For applications where

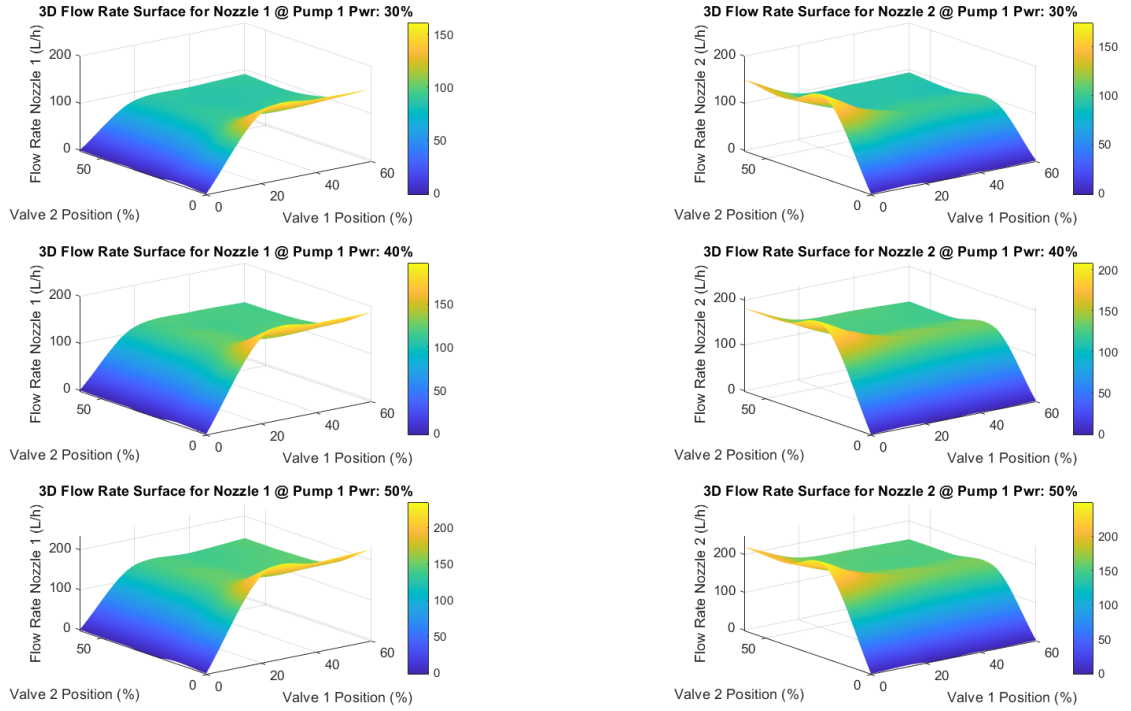


Figure 3.2 – Left-bank open loop response at different pump powers and valve positions.

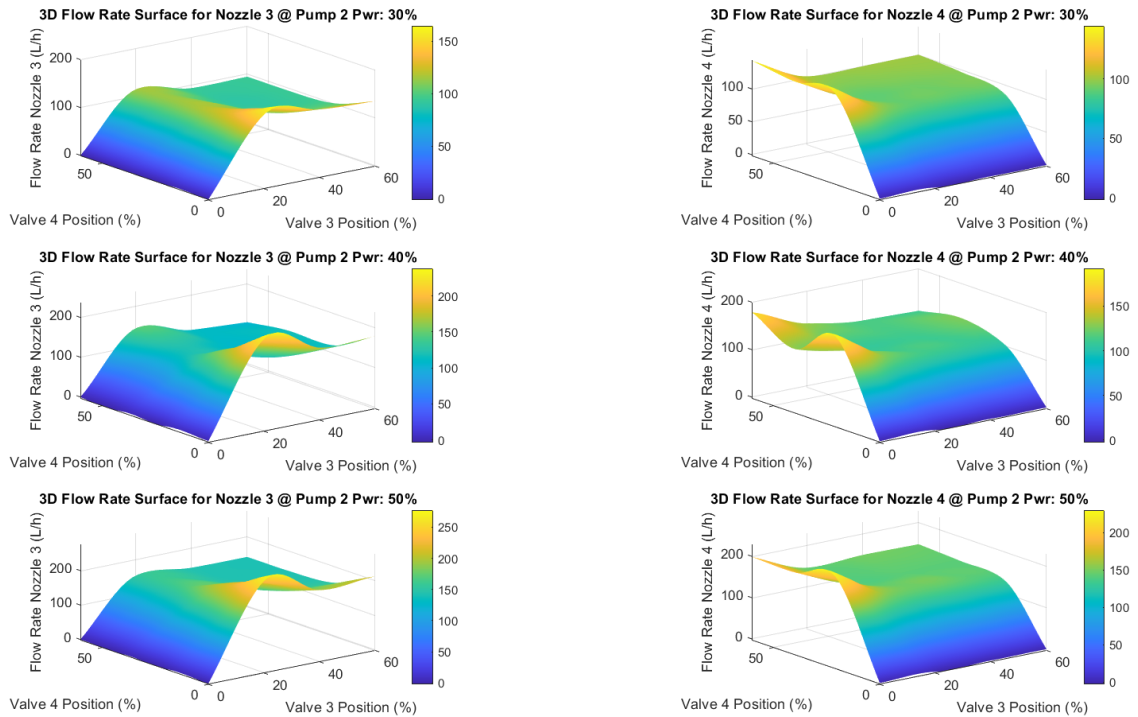


Figure 3.3 – Right-bank open loop response at different pump powers and valve positions.

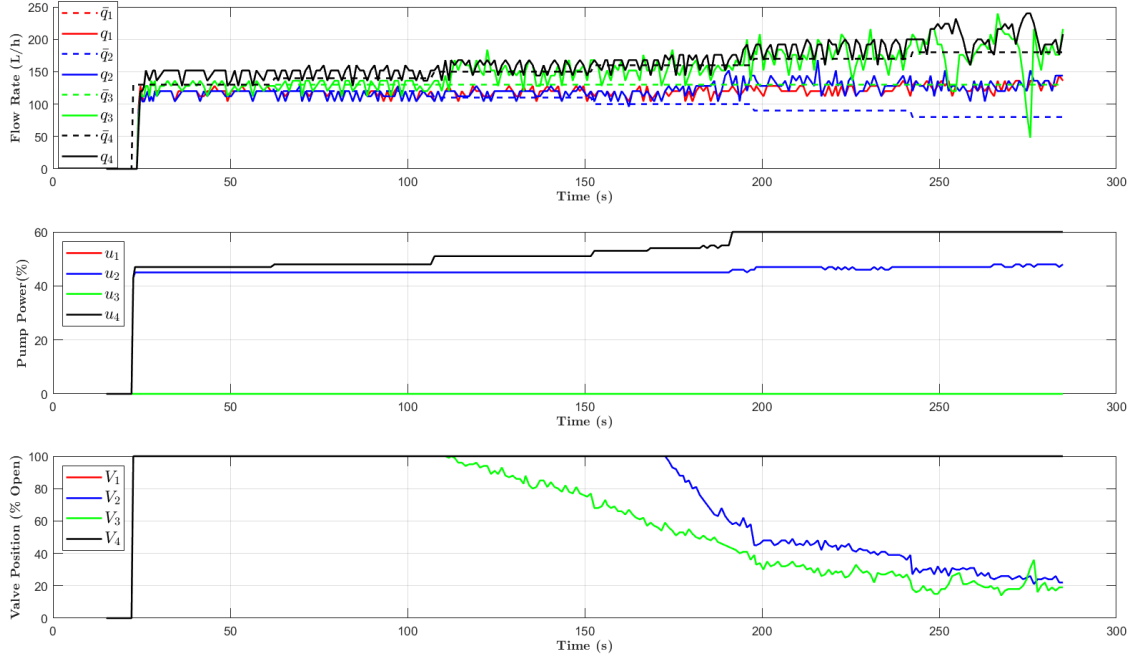


Figure 3.4 – In-Lab Multi-Zone Time Response for banked architecture. The top plot shows the flow rate response to desired step flow rate inputs. The middle plot shows the pump powers computed by the closed-loop pump controller. The bottom plot shows the valve positions computed by the closed-loop valve controller. Note: Pumps 1 and 3 are inactive in the banked system.

precise, zone-specific variable-rate spraying is required, the banked system performance is insufficient. These limitations motivated a transition to the 1-to-1 control architecture, which eliminates nozzle coupling and allows for greater accuracy and independence in meeting prescription demands.

3.2 1-to-1 Architecture

The 1-to-1 architecture assigns a dedicated pump to each nozzle, eliminating the coupling effects present in the banked system and allowing for direct control of individual nozzle outputs. While this configuration increases hardware requirements, it provides a clearer path toward achieving precise, independent nozzle control.

Each nozzle unit has a separate and unique mapping function (M_1 , M_2 , M_3 , and M_4) that converts the pump power to a flow rate.

$$M_j(\hat{u}_j) \rightarrow q_j$$

$$j = 1 \dots 4$$

The inverse mapping function is utilized to calculate the open-loop pump power for the given desired flow rates. Based on the resultant error signals

Table 3.1 – Banked Response Characterization

Nozzle 1		Nozzle 2	
$\Delta\bar{q}$ (L/h)	Steady State Error (%)	$\Delta\bar{q}$ (L/h)	Steady State Error (%)
0	13.9	0	15.1
10	14.9	10	9.8
20	13.9	20	7.7
30	14.5	30	15.9
40	13.4	40	35.5
50	12.7	50	54.5

Nozzle 3		Nozzle 4	
$\Delta\bar{q}$ (L/h)	Steady State Error (%)	$\Delta\bar{q}$ (L/h)	Steady State Error (%)
0	9.4	0	13.4
10	11.1	10	8.9
20	23.3	20	15.6
30	38.0	30	25.4
40	51.2	40	29.4
50	57.7	50	27.8

$$e_j = \bar{q}_j - q_j$$

$$j = 1 \dots 4$$

a proportional-integral controller minimizes the steady-state error. K_p and K_i are the proportional and integral terms of the controllers, respectively.

$$u_j = \hat{u}_j + (K_{p_j} e_j + K_{i_j} \int e_j)$$

$$j = 1 \dots 4$$

3.2.1 1-to-1 Mapping Function

To create the mapping functions for the 1-to-1 architecture, an open-loop test was run for each nozzle unit (Figure 3.6). During the open-loop tests, all four nozzle units were running simultaneously to reflect typical application conditions of the sprayer system. Additionally, in previous research, Zhuta found that for a given pump power, valve position has virtually no effect on the flow rate of the system [10]. Thus, the valve position for each nozzle unit was held constant at 80% open for each test. Five trials were run for each nozzle unit, and the average flow rate at various pump powers was used to generate the mapping functions.

In Figure 3.6, it can be seen that after approximately 47% pump power, the flow rate output begins to plateau or decrease. This is due to power consumption limitations in the sprayer system. Thus, the pump power for each nozzle unit is limited to 45% in the closed-loop 1-to-1 control architecture.

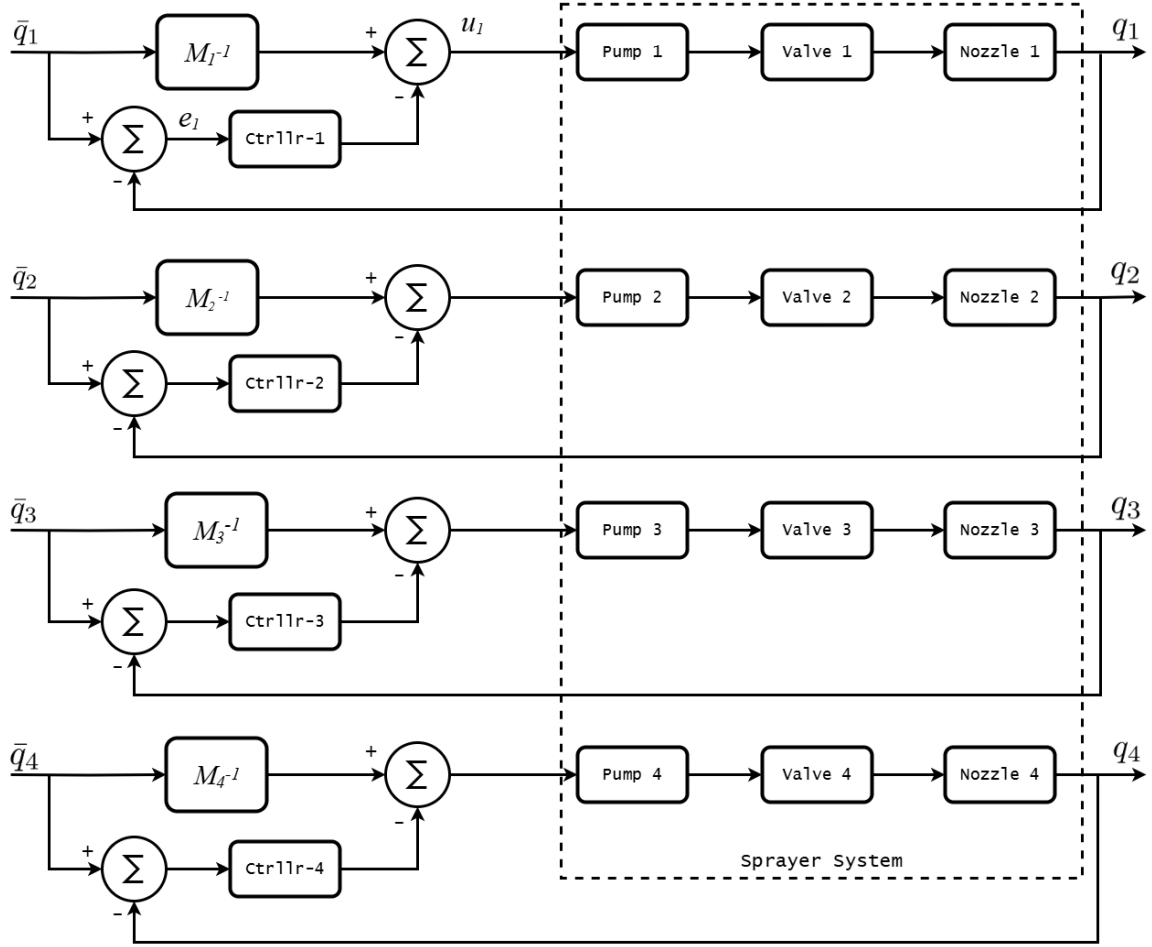


Figure 3.5 – Control Block Diagram Representation of Sprayer 1-to-1 Control System

3.2.2 Testing and Results

3.2.2.1 Sensor Validation

Since the sprayer system utilizes an error-based controller, its performance depends on the accuracy of each flow sensor. Thus, sensor validation testing was performed to determine the accuracy of the flow sensors.

For each nozzle unit, a desired flow rate was set, and for an approximately 30-second window, the dispensed water was collected. After each trial, the average flow rate was calculated by dividing the volume of water collected by the total time of the trial. The calculated average flow rate was compared to the average flow rate measured by the flow sensor. In order to ensure that these measurements were taken at steady-state, the 30-second window began approximately 30 seconds after the desired flow rate was set, to account for the settling time of the closed-loop controller. Each nozzle unit was tested at three different desired flow rates, with five trials conducted per flow rate. Table 3.2 shows the performance of each nozzle. Between 1-3.6% of the system's error can be attributed to sensor accuracy error. This sensor accuracy error can be attributed in part to the resolution of the flow sensors being used. The data recorded from the flow sensors has a resolution of

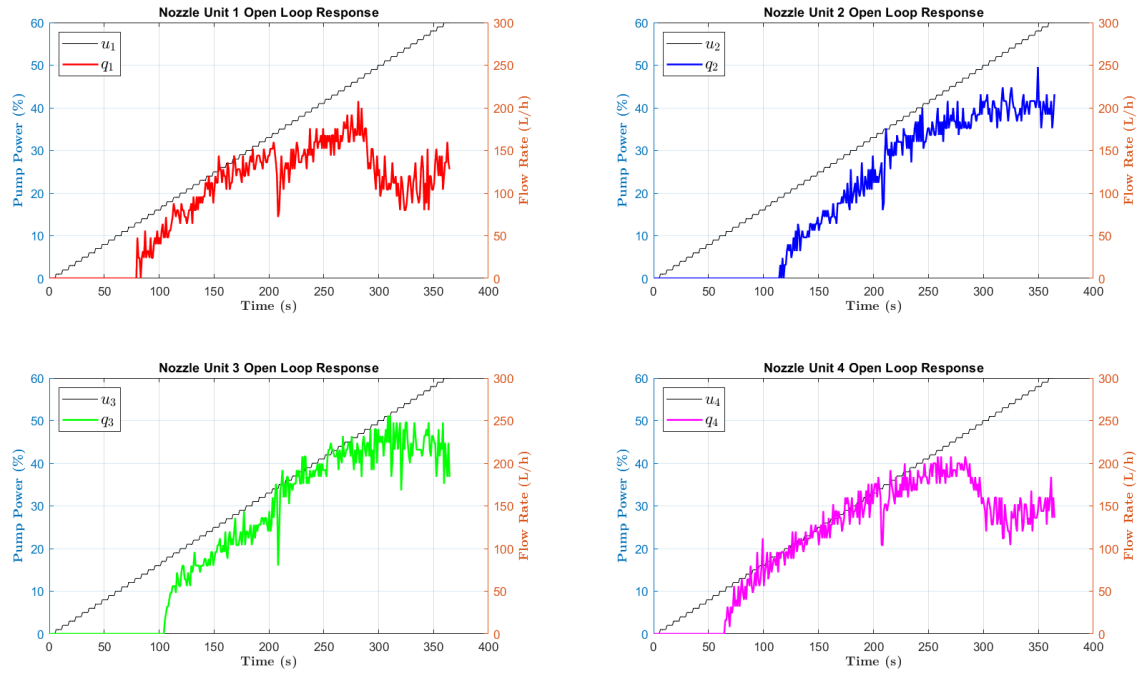


Figure 3.6 – Sprayer system open loop response for each nozzle unit in 1-to-1 architecture. Each valve is set to 80% open.

approximately 8L/h per bit. Thus, Root Mean Squared (RMS) calculations were performed on all data to provide greater insight regarding the steady-state behavior of the system.

Table 3.2 – Nozzle Sensor Accuracy and Validation

Nozzle 1 Flow			Nozzle 2 Flow		
Desired Flow	Sensor Reading	Actual Flow	Desired Flow	Sensor Reading	Actual Flow
121.0	121.2	123.0	121.0	120.4	124.6
154.0	155.1	158.2	154.0	154.2	159.1
169.0	169.4	171.6	169.0	169.1	173.6
Nozzle 3 Flow			Nozzle 4 Flow		
Desired Flow	Sensor Reading	Actual Flow	Desired Flow	Sensor Reading	Actual Flow
121.0	116.6	122.3	121.0	122.1	123.5
154.0	153.3	157.4	154.0	158.2	158.0
169.0	165.5	170.1	169.0	174.7	172.8

In previous research conducted with this sprayer system, Zhuta found that in the field 1-6% of the system error is from sensor accuracy [10]. Since the nozzle units, and thus the flow sensors, remain unchanged from Zhuta's research, and the in-lab sensor validation data corroborate Zhuta's findings, sensor validation in the field was not re-performed.

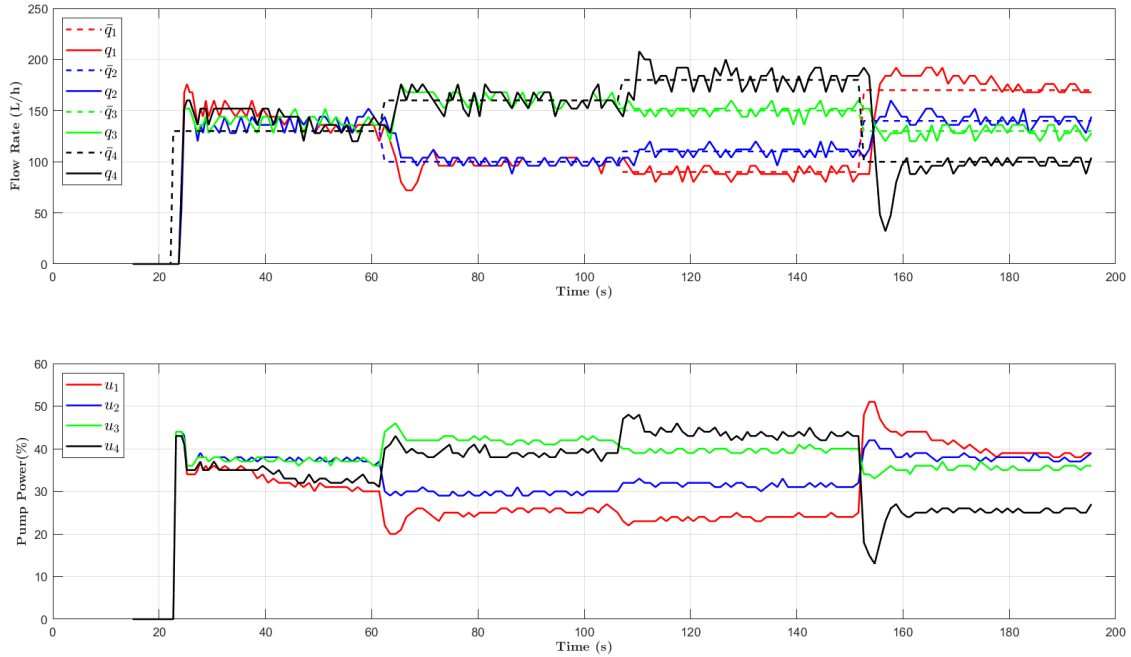


Figure 3.7 – In-Lab Multi-Zone Time Response for 1-to-1 architecture. The top plot shows the flow rate response to desired step flow rate inputs. The bottom plot shows the pump powers computed by the closed-loop controller.

Table 3.3 – Desired Flow Rates for Multi-Zone Test

Zone	Nozzle 1	Nozzle 2	Nozzle 3	Nozzle 4
1	130.0	130.0	130.0	130.0
2	100.0	100.0	160.0	160.0
3	90.0	110.0	150.0	180.0
4	170.0	140.0	130.0	100.0

3.2.2.2 In Lab Testing

The in-lab closed-loop tests for the 1-to-1 architecture demonstrate that the controller was able to achieve independent nozzle control with acceptable performance across all four nozzles. Table 3.4 summarizes the average settling times and steady-state errors for each nozzle, while Figure 3.7 illustrates the measured flow rate responses and the corresponding pump power outputs for four different zones of desired flow rates. Table 3.3 shows the desired flow rates for each zone.

The average settling time ranged from 3.7 seconds to 9.7 seconds. This difference can be attributed to minor differences in pump dynamics and sensor noise among the different nozzle units. Nonetheless, all nozzles converged within 10 seconds of a step change, which is suitable for agricultural spraying applications where time constants are relatively slow compared to the rover's speed.

Steady-state error across the nozzles ranged from 5.3% to 6.1%. When accounting for the 1-3.6% error attributed to sensor resolution, the effective control error is approximately

Table 3.4 – In Lab Multi Zone Response

	Nozzle 1	Nozzle 2	Nozzle 3	Nozzle 4
Average Settling Time (seconds)	9.7	5.8	3.7	7.6
Average Steady State Error (%)	5.4	5.6	5.3	6.1

1.5-5%. This indicates that the PI controller, in conjunction with the feedforward mapping functions, is able to maintain close tracking of the desired setpoints. These results represent a clear improvement from the banked system, where nozzle coupling introduced steady-state errors greater than 10% under many operating conditions. Overall, the in-lab testing validates the 1-to-1 control architecture achieves robust and independent control of each nozzle unit.

3.2.2.3 Field Testing

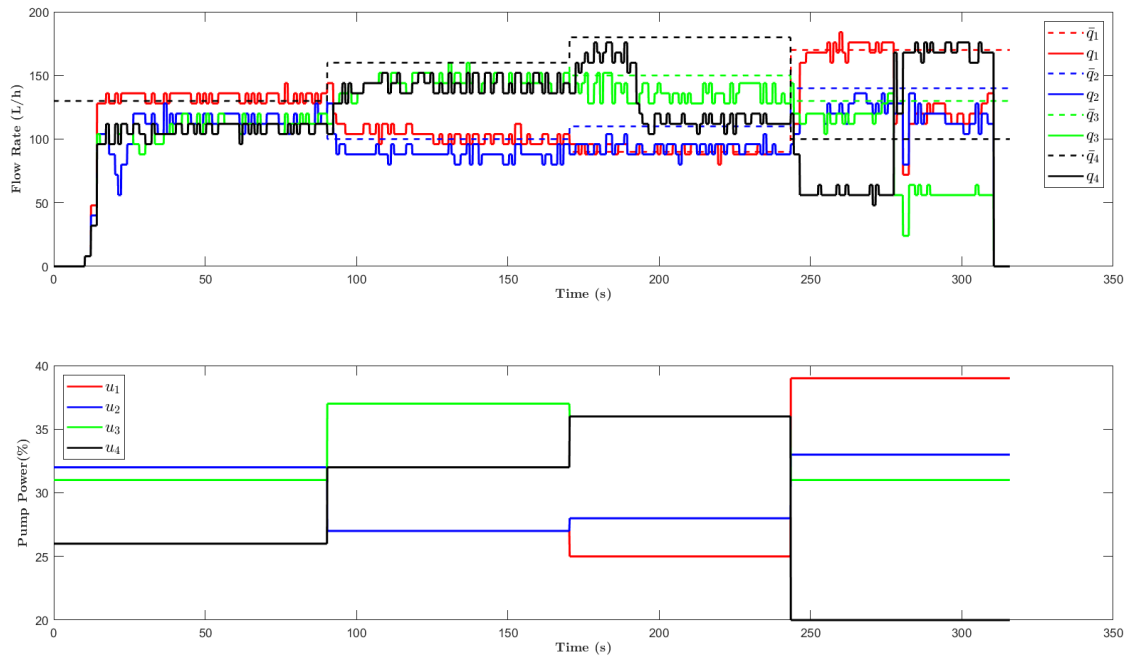


Figure 3.8 – Multi-Zone Time Response in the field. The top plot shows the flow rate response to desired step flow rate inputs. The bottom plot shows the pump powers computed by the closed-loop controller.

Field testing was performed on a nearby farm to simulate environmental conditions—wind, dust, and uneven terrain—that the rover and sprayer system would face during its operation. Testing occurred while the rover was moving at speeds ranging from 0.2 to 0.4 meters per second. In each test, the Agbot drove for 80 meters. This distance was chosen based on the approximate length of a row at the farm. Figure 3.8 shows a time history response for the sprayer system in the field as it traverses through four different zones, each 20 meters long, with the same desired flow rates as tested in the lab (Table 3.3).

Figure 3.8 and Table 3.5 showcase the response of the sprayer system in the field. It can be seen that the nozzle units have settling times ranging from 9.2 seconds to 20.7 seconds. Nozzles 3 and 4, in particular, required nearly twice as long as in the laboratory setting to reach steady state. This difference is likely due to supply-side disturbances, such as the inlet tubing not being fully submerged as water levels dropped, as well as environmental disturbances, including wind and rover vibrations. These conditions also contributed to larger steady-state errors in the field, with values ranging from 9.1% to 28.9%.

Despite these limitations, the controller maintained the ability to track zone-based prescription commands in the field. The 1-to-1 control architecture outperformed the banked configuration by maintaining nozzle independence and adapting to zone transitions as the rover moved across the test field.

Table 3.5 – Field Multi Zone Response

	Nozzle 1	Nozzle 2	Nozzle 3	Nozzle 4
Average Settling Time (seconds)	9.2	14.2	18.7	20.7
Average Steady State Error (%)	10.8	9.1	13.9	28.9

Chapter 4

Prescription Model

In the Agbot sprayer system, the prescription model forms the bridge between the high-level agricultural goals and the low-level control architecture of the pumps and valves. It determines the desired flow rates for each nozzle based on three main inputs: rover position, rover speed, and row width.

Prescription models can be built from a variety of data sources, including soil maps, yield maps, sensor networks, or remote sensing. These data sources are used to create prescription maps or zone-based requirements, which are then converted into real-time control setpoints. These setpoints account for the following operating conditions of the rover:

1. **Position:** determines which zone-specific prescription should be applied
2. **Speed:** scales flow rates so application volume per unit area remains constant
3. **Row Width:** distributes application rates across active nozzles.

For this sprayer system, three levels of prescription complexity were defined: basic, simple, and complex. These levels balance implementation simplicity with agronomic precision. Figure 4.1 illustrates these models. For all three models, the row width is assumed to be constant.

The basic prescription model has a constant application rate, regardless of the rover's location. The desired flow rate is based solely on rover speed

$$\bar{q} = PvW$$

where P is the prescription rate (L/m^2), V is the rover velocity (m/s), and W is the row width (m). The prescription rate, P , represents the application requirement, in L/m^2 , needed to provide the desired volume per area of fluid at the rover's current velocity. The basic prescription model is robust and does not require prescription maps, making it easy to implement computationally. However, it cannot adapt to field variability and may lead to over- or under-application in the field since it does not leverage the full potential of data-driven farming.

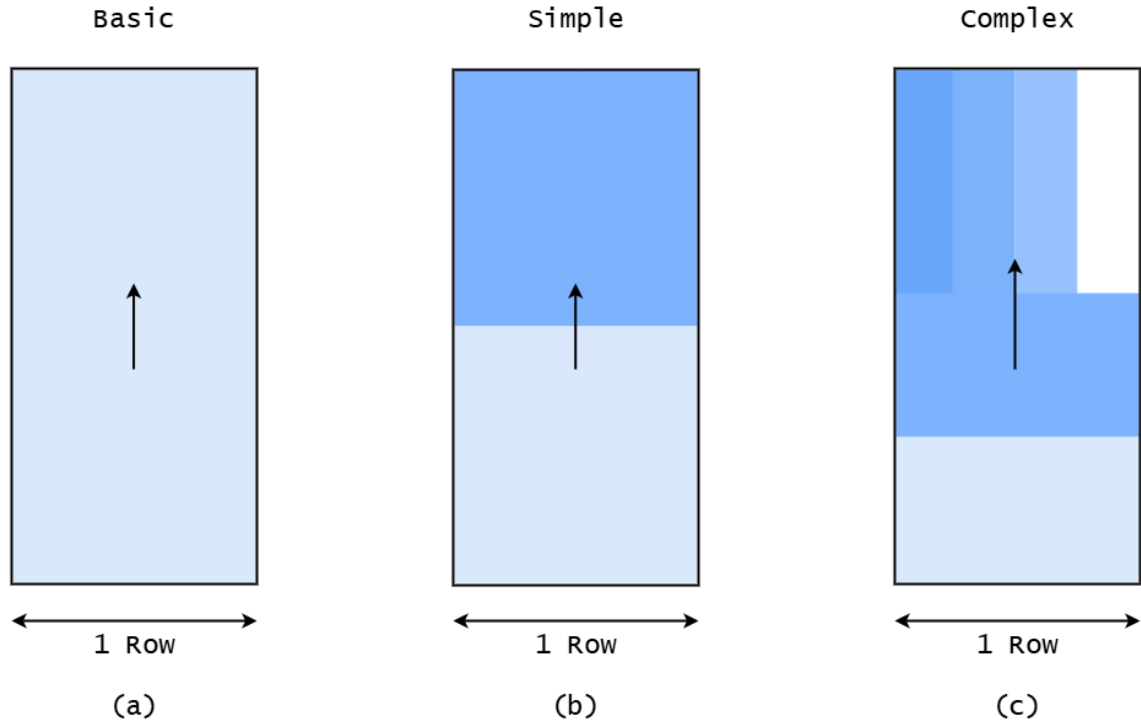


Figure 4.1 – Prescription Model Levels of Complexity. (a) Basic prescription with constant volume per area. (b) Simple prescription with different volumes per area based on longitudinal position. (c) Complex prescription with different volumes per area longitudinally and laterally.

The simple prescription model allows the application rate to vary longitudinally, and thus depends on the rover's position along its path.

$$\bar{q} = P(pos)vW$$

In this model, the prescription rate is position-dependent, enabling zone-based spraying. The simple model utilizes simple prescription maps and GPS to gain spatial adaptability, but still has a coarse application.

In the complex prescription model, the application rate can vary both longitudinally and laterally across the row's width, ultimately enabling nozzle-level precision.

$$\bar{q} = \sum_{i=1}^n P_i(pos)v \frac{W}{n}$$

where $P_i(pos)$ is the prescription at the lateral location of nozzle i and n is the number of nozzles. This complex model allows for fine-grain control applications like row-specific spraying or plant-level targeting; however, it has the highest data demand of the three models.

Due to the benefits and trade-offs of each model level, they each have a unique best use case. The basic model is ideal for uniform fields and baseline testing that does not depend

on location. The simple model is great for fields with moderate variability and only row-level differences. The complex prescription model is best for farms with high-value crops, like vineyards and orchards.

4.1 Prescription Model Validation

To validate the prescription model, both the simple and complex models were tested in the field. A zone-based prescription model was used in combination with the Agbot's GPS and speed data. As the Agbot travels down a row in the field, its GPS data is used to determine which of the four zones it is currently in. The zone corresponds to a set of four desired flow rates which are then sent from the Jetson on the main rover to the Arduino Mega 2560 on the sprayer system.

In order to evaluate the prescription model, the relationship between volume dispensed and distance traveled by the rover was analyzed. Figure 4.2 shows the instantaneous measured volume dispensed by each nozzle compared to the desired instantaneous volume calculated from the prescription model. The measured volume dispensed plot shows expected variability due to flow sensor noise, transient dynamics, and supply-side and environmental disturbances. The desired volume dispensed plot shows more consistent step changes, corresponding to the different prescription zones encountered as the rover traveled through the field. There is minor noise on the desired plot due to differences in irregular sampling times in the collected data. Table 4.1 quantifies this comparison by reporting the RMS error for each nozzle across four zones. The data highlights that performance varied both by nozzle and by zone. For instance, Nozzle 1 maintained relatively low errors in Zones 1–3 (0.9–5.1%) but showed a larger deviation in Zone 4 (16.3%), while Nozzle 3 exhibited consistently higher errors, peaking at 32.2% in Zone 4. These differences reflect the combined influence of sensor accuracy, nozzle dynamics, and environmental disturbances in the field.

Figure 4.3 shows the cumulative volume dispensed by each nozzle as a function of distance. This representation smooths out instantaneous fluctuations and highlights how closely the system adheres to the prescription model over the entire test path. All nozzles show near-linear cumulative growth with distance, which is consistent with the model's goal of constant volume per area application. In addition, differences in slope can be seen, which reflect zone-specific prescriptions and variability at the nozzle level.

Table 4.1 – Prescription Model Response: Average Root Mean Square (RMS) Error (%)

Zone	Nozzle 1	Nozzle 2	Nozzle 3	Nozzle 4
1	5.1	20.0	22.4	23.6
2	4.0	10.6	9.9	11.8
3	0.9	15.3	9.0	28.4
4	16.3	15.3	32.2	12.5

The prescription model serves as the bridge between the field prescription maps and real-time sprayer control, enabling scalable precision spraying from uniform treatment to site-specific applications.

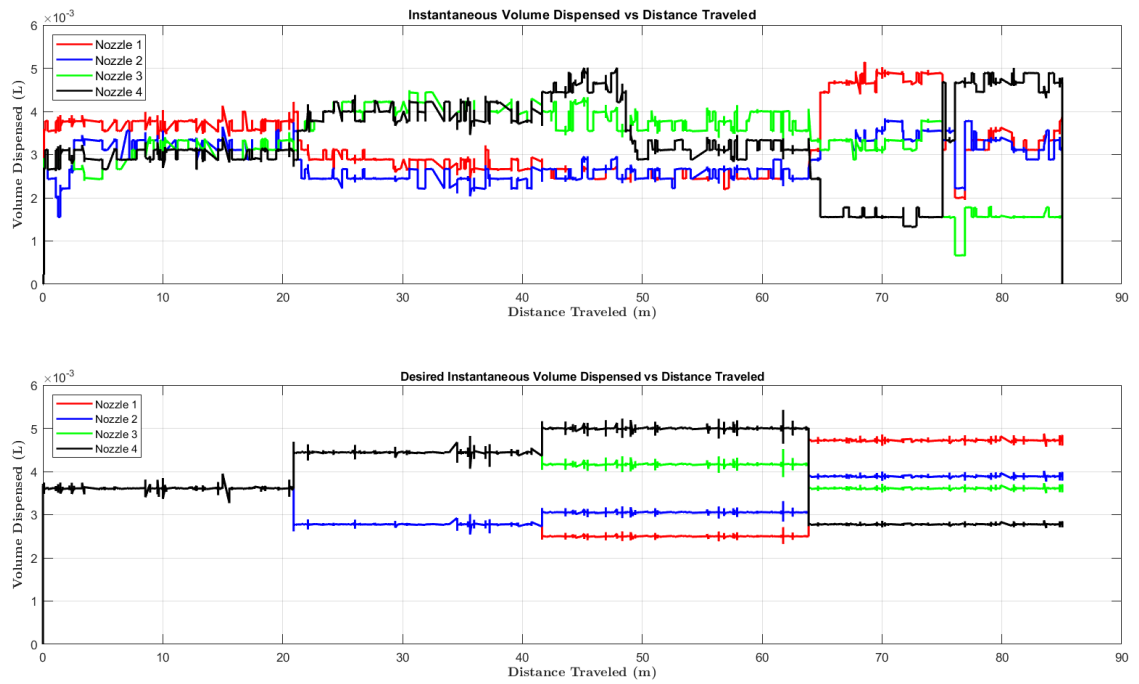


Figure 4.2 – Instantaneous volume dispensed as the Agbot traverses 80 meters. Top plot shows volume dispensed in the field. Bottom plot shows desired volume dispensed.

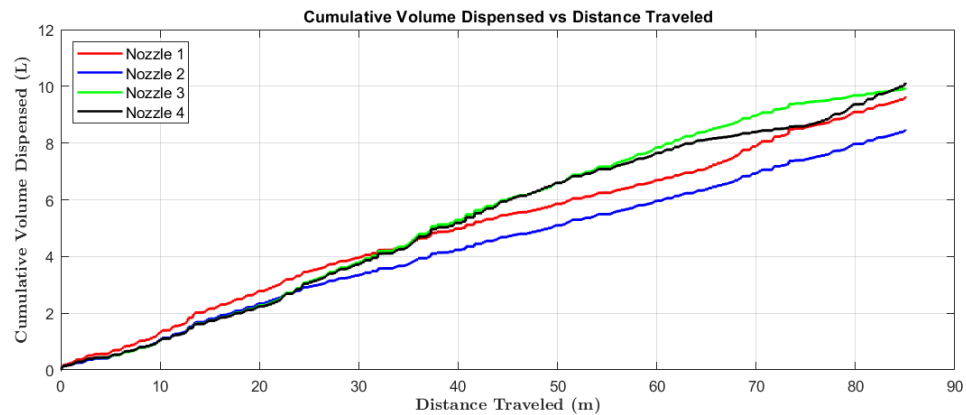


Figure 4.3 – Cumulative volume dispensed by each nozzle as Agbot travels 80 meters.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

In this capstone, I have demonstrated that independent nozzle control can be achieved on a modular sprayer system when utilizing a 1-to-1 pump configuration, which overcomes the limitations of a banked configuration. The sprayer system uses a closed-loop proportional integral controller with a feedforward mapping function to dispense flow rate with an error margin of 5.3% to 6.1% in the lab and 9.1% to 28.9% in the field.

Additionally, a prescription model was developed to integrate rover position and speed with nozzle flow rate commands, bridging agronomic intent with engineering implementation. Experimental validation using volume-distance plots confirmed that the prescribed application rates were met across the rover's path. RMS errors varied across nozzles and zones—from strong tracking in zone 3 for nozzle 1 (0.9% error) to larger deviations in zone 4 for nozzle 3 (32.2% error). This spread highlights both the capability of the model to achieve zone-based spraying and the challenges posed by supply-side and environmental disturbances and nozzle-specific dynamics. This capstone also outlined three levels of prescription complexity, which establishes a scalable framework that is adaptable to diverse agricultural use cases.

Overall, this research contributes a validated prescription model and control framework and is a step toward fully automated, sustainable precision agriculture.

5.2 Future Work

Future work could focus on further integration of the sprayer system with the Agbot rover. One area of development is testing the system in more realistic agricultural environments, such as in furrows, where the rover must traverse different rows and execute turns at row ends. Additionally, end-to-end automation could be enhanced by enabling the rover to autonomously dock and refill its water supply when levels run low, allowing for extended, uninterrupted operation in the field.

Another opportunity for future improvement lies in advancing the prescription model from its current open-loop implementation to a closed-loop framework. Currently, the system dispenses fluid according to the prescription model and employs closed-loop control

for flow rate, but it does not provide feedback at the ground level regarding actual volume dispensed per area. Exploring closed-loop prescription control—where rover speed and real-time flow rate data are combined to estimate actual application volume—could enable adaptive corrections to ensure that field-level prescriptions are consistently met. This extension would close the loop between high-level prescription and ground-level application, supporting more reliable and precise delivery of agricultural inputs.

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